#4 · 10.5% of marks

Ch 8 — Mutual Funds

Tested every attempt: NAV working, holding-period return (payout vs reinvest), dividend vs bonus plans, back-solving issue price, closed-end premium / discount.

1 Set-up & structure

- → *MF = **trust** pooling savings via **units**; invests in capital & money market; income & gains shared in proportion to units*
- → *Four constituents: **Sponsor** → **Trust (trustees)** → **AMC** → **Custodian**, all SEBI-registered. Trust deed registered under the Registration Act, 1908*
- → *Sponsor: **5 years** in financial services, holds **≥ 40%** of AMC net worth. AMC net worth **₹100 crore** (SM). Trustees ⅔ **independent**; AMC board **50% independent**. AMC removed by majority of trustees or **75%** of unit holders*
- → *Custodian: SEBI-registered, **independent of sponsor & AMC** (DP for demat)*

2 Classification

Basis	Types
Structure	***Open-ended** (buy / redeem any time with fund; no maturity) · **Close-ended** (NFO then exchange; exit on exchange / maturity / repurchase window) · **Interval** (need not list; in-built redemption window; can issue fresh units at NAV in the interval; no fixed maturity)*
Asset	*Equity (**growth** — long-term appreciation; **aggressive** — IPOs, speculative; **income** — high-dividend stocks) · Debt / bond (risks: **I-C-P-R** — interest rate, credit, prepayment, reinvestment) · Money market / liquid · Gilt · Hybrid / balanced*
Special	***Index** · **International** (raises in India, invests abroad) vs **Offshore** (outside India, raises globally to invest in India) · **Capital protection oriented** (protection from portfolio structure, not guarantee; close-ended, listed, structure rated) · **Quant** (rule-based models; no bias, low cost; relies on history) · **Fund of funds** · **ETF** · **Arbitrage** (spot vs futures) · **Sector** vs **thematic***
Ownership	***Public sector** (sponsored by a public-sector company / bank) · **Private sector** · **Foreign** (foreign company operating from India, raises and invests in India)*
Diversified equity (SM 3.2)	***Flexicap** · **Multicap** · **Contra** (out-of-favour / undervalued companies; blends growth + value; needs aggressive risk appetite) · **Index** · **Dividend yield** (yield above the index's; less volatile than growth stocks)*
Others (SM 3)	***Thematic** — broader than sector (multi-sector, international, commodity); market may take long to recognise the theme · **Arbitrage** — open-ended, low-volatility, cash equities + derivatives + debt; spot vs futures differential · **Cash fund** — open-ended liquid, very short-dated debt & money market, growth & dividend options*
Schemes	***Flexicap** (any proportion) vs **Multicap** (≥ 25% each large / mid / small) · **FMP** (no interest-rate risk, still credit risk; close-ended, illiquid) · **ELSS** (80C, 3-yr lock-in; SIP preferred)*
Options	***Growth** (nothing paid) · **Dividend payout** (NAV falls by dividend) · **Dividend reinvestment** (extra units at ex-dividend NAV)*
Plans	***Direct plan** (from 1 Jan 2013) — no distribution expense / trail → lower TER, **higher NAV** than regular*

⚠ TRAP

*Interval ≠ close-ended. CPOF ≠ capital **guaranteed**. FMP is free of **interest-rate** risk, **not** credit risk.*

3 ETF vs index fund; hedge fund vs MF

ETF	Listed; trades all day at real-time prices close to NAV (open-ended fund deals once a day at NAV); **authorised participants** act as market makers; low cost, tax-efficient, no paperwork. Types: index, commodity (gold), bond, currency
Hedge fund vs MF	Lightly regulated private vehicle for selected clients; little public disclosure; **performance fee** (losses borne by investors); uses **leverage, long / short, derivatives**

High-water mark fee (SM: ANP Plan): ₹20 cr; 0.10% fixed + 2% above ₹21 cr. Return 29% → value 25.8 cr → fee ₹2 L + 2% × 4.8 cr = **₹11.60 L**; return 4.5% → 20.9 cr < 21 → **₹2 L only**. Trap: fee on excess over HWM, not on the year's gain.

4 Advantages (12) & drawbacks (4)

Advantages

Professional management · diversification · convenient administration · higher returns · **low cost** (max **2.50%** for equity funds; excess borne by AMC) · liquidity · transparency · other benefits (SIP / SWP / switch) · highly regulated · economies of scale · flexibility · convenience

Drawbacks

1. No guarantee of return (under-performance, entry / exit timing, crashes) · 2. Diversification caps upside · 3. Picking the right fund is hard; past performance no guide · 4. Costs; AMC fees not linked to performance

5 Key terms

Term	Meaning / formula
NAV	Net assets ÷ units outstanding. Net assets = MV of investments + receivables + accrued income + other assets − accrued expenses − payables − other liabilities
Loads	Public offer price = NAV ÷ (1 − entry load) · Redemption = NAV ÷ (1 + exit load) [or NAV × (1 − exit load) where the question says]. Exit load goes back to the **scheme**. No entry load now
Required MF return	$r_2 = r_1 \div (1 - \text{initial expenses}) + \text{recurring expenses}$ (10%, 6%, 2% → **12.64%**)
Expense ratio (TER)	Mgmt & advisory fees, travel, audit, custodian, registrar — **excludes brokerage**; charged daily (1/365) before NAV
Trail commission	0.30–0.75% p.a. on daily average AUM, paid monthly, while money stays invested
Side pocketing	Stressed / downgraded assets moved to a segregated portfolio; main NAV reflects liquid assets; protects continuing holders (IL&FS). Recovery paid only to holders on segregation date
Tracking error	$\sqrt{[\Sigma(d - \bar{d})^2 \div (n - 1)]}$, d = fund − benchmark return. Causes: transaction cost, AMC fees, expenses, cash holdings, sampling. Higher TE = higher risk even if outperforming
Dividend equalisation	New investor pays / exiting investor gets undistributed income per unit: issue = NAV + entry load + income p.u.; repurchase = NAV − exit load + income p.u.

6 Return formulas

HPR = $(NAV_1 - NAV_0 + \text{income distributed} + \text{capital-gain distributed}) \div NAV_0$

Reinvestment: extra units = distribution ÷ reinvestment NAV; return = (total units × closing NAV − cost) ÷ cost

Effective yield p.a. = [(units × closing NAV − investment) + dividends] ÷ investment × **365 ÷ days**

Annualise monthly return × 12 (simple) unless told to compound; **real return** = (1 + nominal) ÷ (1 + inflation) − 1

Pattern	Method
Payout vs reinvest (Sep 2025 Q3(b))	Compute both; reinvest at the given (average) NAV; pick higher return
Entry load + appreciation + distribution reinvested + exit load + tax + inflation (Sep 2025 Q5(b))	Units = amount ÷ POP; end value = units × NAV × (1 − exit load); tax on gain; real return
Dividend % declared	Dividend = units × **face value ₹10** × % (not × NAV)
Find month-end NAV from annual return (PYQ May 2024)	Monthly = 16% ÷ 12 = 1.33%; 0.0133 = (NAV − 78.50 + 0.70) ÷ 78.50 → **₹78.84**
Days invested from yields	Days = period yield × 365 ÷ yield p.a.; count back

7 Dividend vs bonus vs growth plans

→ Build a **units table** per plan to redemption; value = units × closing NAV; deduct STT / tax on short-term units only; annual return = gain ÷ investment × 12 ÷ months

→ **Back-solve issue price** (Jan 2026 Q1(a)): value now = investment × (1 + yield × years); units now = value ÷ NAV; **undo bonuses latest first** (× b ÷ (a + b)); issue price = investment ÷ original units. SM: ₹16 L, 17.5% × 5 → ₹30 L ÷ ₹100 = 30,000 → 24,000 → 20,000 → **₹80**

8 NAV statement (new fund, one month)

1. Cash = amount raised − underwriting commission − marketing / issue costs − investments bought + sale proceeds + dividends received − expenses **paid**
2. Net assets = cash + market value of holdings + receivables (uncollected dividend) − **arrears / outstanding expenses**
3. Distribution = % of **realised** earnings only (sale gains + dividends), deducted from cash
4. NAV per unit = net assets ÷ units (underwritten issue → all units outstanding)

⚠ TRAP

Only expenses **paid** leave cash; arrears are deducted from net assets. Units issued on a cheque = cheque ÷ **unrounded** current NAV; NAV unchanged that day.

HIGH

9 Evaluating a mutual fund

Quantitative (5)

1. Risk-adjusted returns · 2. Benchmark returns (alpha) · 3. Comparison with peers · 4. Returns across economic & market cycles · 5. Financial measures — expense ratio, Sharpe, Treynor, Sortino

Qualitative (3)

1. Quality of portfolio (blue chips, diversification, style; debt: credit quality, maturity, modified duration; low turnover) · 2. Track record & competence of fund manager · 3. Credibility of fund house team

Measure	Formula / note
Sharpe	$(R_p - R_f) \div \sigma_p$ — total risk
Treynor	$(R_p - R_f) \div \beta$ — systematic risk
Sortino	SM: variation of Sharpe using **downside deviation** instead of total σ in the denominator
Jensen α	$R_p - [R_f + \beta(R_m - R_f)]$
M^2 (not in SM)	$R_f + \text{Sharpe}_p \times \sigma_m$ — compare with R_m
Information ratio (not in SM)	$(R_p - R_{benchmark}) \div$ tracking error

10 Reverse-engineering β and NAV

Given	Get
Sharpe & σ → excess return; with Treynor	**β = (Sharpe × σ) ÷ Treynor**
ρ, σ_p, market variance	$\beta = \rho\sigma_p \div \sigma_m$ ("variance 9%" → $\sigma_m = 3$)
Sharpe, Treynor, σ_m	σ = excess ÷ Sharpe; β = excess ÷ Treynor; $\rho = \beta\sigma_m \div \sigma$; **$R^2 = \rho^2$**
Jensen α & Treynor	CAPM return = actual − α; solve R_f and β simultaneously

Expected NAV (SM): NAV ₹70.71, 99% equity, Sharpe 2, σ 11.25, Treynor 15 → β = 22.5 ÷ 15 = 1.5. Market −5% → equity ₹70 × (1 − 7.5%) = 64.75; cash 0.71 − monthly cost 0.25 = 0.46 → **NAV ₹65.21**. Trap: **don't apply β to the cash part**.

11 Closed-ended fund premium / discount

Market price = NAV × (1 ± premium / discount). Investor's return uses **market prices**, so it beats NAV growth if the discount narrows. PYQ May 2026: geometric mean return 3.771% vs NAV growth 2.76%. Geometric mean = $[(1 + r_1)(1 + r_2)\ldots]^{1/n} - 1$.

12 Role of fund manager & FIIs

- → **Active** fund: outperform, generate positive alpha via research & stock picking. **Passive**: match index with minimum tracking error
- → Other roles (**C-M-W-O**): **C**ompliances (SEBI / AMFI, reporting), constant **M**onitoring of performance, **W**ealth creation & protection, control over **O**utsourced work
- → **FIIs** buy units directly from the issuer or through a registered broker on a recognised exchange within SEBI limits; benefits: **better corporate governance**, **more competition**, **capital inflow** (jobs & forex)
- → Fund hedge with index futures: contracts = β × equity value ÷ (futures × lot) — **cash not hedged** (₹90 cr, β 1.1, 4,300 × 50 → 4,605)

13 Formula plans — constant ratio plan

- → Total value = bonds + units × NAV; when NAV move hits the trigger, **reset to the fixed ratio** — buy units after a fall, sell after a rise
- → ICAI checks each NAV against the **previous** one (32 → 38 = +18.75% triggers)
- → Wins in **choppy** markets ending near the start (sells high, buys low); loses in a strong one-way rise (keeps selling the winner)

14 SIP / STP / SWP & vocabulary

- → SIP: fixed amount in; average cost = total ₹ ÷ total units = **harmonic mean** of NAVs ≤ average NAV (₹10,000 at 20, 16, 25, 20 → ₹19.75 vs ₹20.25)
- → SWP: fixed amount out · STP: fixed amount moved scheme to scheme
- → NFO · AUM · folio · **cut-off time** (before → that day's NAV; after → next day's)
- → Gilt fund: rates expected to fall → **lengthen duration**; to rise → shorten

15 Correlation — how Ch 8 actually connects

Ch 8 idea	The actual link (app-written cross-reference; the Ch 8 side is SM)
Sharpe / Treynor / Jensen	**Ch 6**: same formulas; Sharpe for an undiversified investor (σ), Treynor for a diversified one (β), Jensen α positive = fund beat CAPM.
Debt / gilt fund NAV	**Ch 5**: rates ↑ → bond prices ↓ → NAV ↓; % change ≈ −modified duration × Δy, so longer-duration funds swing more.
Hedging a fund	**Ch 9**: sell index futures = Value × β ÷ (futures price × lot) to protect NAV against a market fall.
Index funds / ETFs	**Ch 6** passive strategy and **Ch 4** efficient-market logic — low cost, tracks index.
Units held by a company	**FR Ind AS 109**: MF units generally fail SPPI and are not equity instruments for the OCI election → FVTPL.

> ✍ **Write in exam**
> "NAV per unit = (market value of investments + receivables + accrued income + other assets − accrued expenses − payables − other liabilities) ÷ number of units outstanding."